

Practice Guide 3: Breaking the Filters (Counter Examples and Useful Examples)

Aprendizagem Aplicada à Segurança

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Objective

You play the **attacker** against the filters of the previous guides, at two levels, and then you defend.

- **Level 1, counter example:** find a pattern that bypasses **Naive Bayes, logistic regression** and an **embedding model**. You need not keep the message working; you need to *name the blind spot*.
- **Level 2, useful example:** find a rewrite that still works as a scam: the **payload** (phone numbers and short codes, links, amounts, call to action) is intact, the **meaning** is kept (an independent embedder says so), and it **looks normal** to a reader. This is what a scam campaign could reuse.
- **Defend:** normalise the input, retrain on the attack, and measure what the attacker does next.

Lab use only. You attack the models you trained yourself, on the public SMS corpus. Before writing code, state the **threat model**.

Axis	This lab
Goal	a spam message is classified as ham (<i>evasion</i>)
Capability	append words, replace words, insert characters; level 2: the payload must stay intact
Knowledge	white-box (weights) or black-box (a score per message: each score is a <i>query</i>)
Constraint	level 2: payload, meaning and a normal look

- **Notebook to complete:** `guide_02_03.ipynb`. **Time:** about 2 hours (suggested: A 10 min, B 30, C 20, D 35, E 25). **Prerequisites:** Guides 1 and 2 and the lecture of class 4.

- **Requirements:** make venv. You also need the llama.cpp servers (make llama-up, see Guide 2): the MiniLM server is the *judge of meaning* in part D, the Nomic server is an optional third victim (AAS_SLM=skip leaves the Nomic victim out). Set AAS_W2V=skip to skip the Google News download in part D (no synonym operator then).

Background

Additive scores. For Multinomial Naive Bayes the log-odds of a message are exactly $z = \log \frac{P(\text{spam})}{P(\text{ham})} + \sum_w c_w \text{llr}_w$, with $\text{llr}_w = \log \frac{P(w|\text{spam})}{P(w|\text{ham})}$: every word adds its own evidence, so the cheapest attack is a *sort*: add the words with the most negative llr.

Black-box search. The attacker cannot read the model; they *evaluate* it. Greedy search: append every candidate word, keep the one that lowers the score most, stop when the score is negative. With m candidates and s steps the cost is about $1 + m \cdot s$ **queries**.

Dilution. TF-IDF vectors are L2-normalised and embeddings are averaged over tokens: benign text makes the spam a smaller share of the input.

Level 2. Minimise the change subject to: the filter says *ham*, the **payload** is unchanged, the **visible** text stays close to the original (cosine of an independent embedder $\geq \tau$), and at most B words differ *for a human* (Cyrillic look-alike letters and zero-width spaces are invisible, so they cost nothing). Allowed edits look normal: invisible characters, real benign sentences, near-synonyms, paraphrases; junk words are not.

Defences and the arms race. *Normalisation* (NFKC, remove invisible characters, map look-alikes back) closes the invisible attacks. *Adversarial training* covers the attacks it has seen; the attacker adapts. *Removing the score* (label-only API) and *limiting queries* raise the attacker's price. Each defence has a cost and an answer.

Tasks

Part	What you do	Check
A	Threat model, victims and the public candidate words (given)	
B	NB: reproduce the score from the log-odds, closed-form attack; implement the greedy black-box search ; ASR curves	assert on the score, Questions B1 to B3
C	Dilution; transfer of the crafted messages to the other victims	table, Questions C1, C2

Part	What you do	Check
D	Implement the payload check with regular expressions; run a constrained search (camouflage, synonyms, invisible characters) and measure the useful ASR	assert on the payload check, Questions D1 to D3
E	Implement normalisation; adversarial training and re-attack	assert, Questions E1 to E3

Tips. Every attack is an optimisation problem: gradients when the model is differentiable and known, derivative-free search when it is not. Count queries. Print the adversarial messages: what a model accepts and what a person accepts are different questions.

Questions for the report

1. **(B1)** What are `lt` and `gt`, why are they so “hammy”, and what does that say about the data and about the attacker’s cost?
2. **(B2)** At budget 0 some messages already evade. What are they, and how does this change the meaning of “attack success rate”?
3. **(B3)** Compare the white-box closed form with the black-box curve for Naive Bayes. Which knowledge is worth more, and how many queries did the black-box attacker pay?
4. **(C1)** Which victim resists dilution best? Explain with the way each model turns a message into a score.
5. **(C2)** Do the messages crafted against one victim also evade the others? Why (not)?
6. **(D1)** Compare the useful ASR with the level 1 evasion of Part B. What is the *price of usefulness*, and which constraint costs most?
7. **(D2)** Look at the example message: what did the attacker change, and why would a recipient not notice?
8. **(D3)** Which scam family would you expect to be easiest to disguise, and why? Test it on your messages (`sl.spam_families`).
9. **(E1)** What does normalisation cost, and which of the attacks of Parts B to D does it not touch?
10. **(E2)** Report the clean F1 and the evasion before and after adversarial training. Why does the attack not go to zero, and what would you try next as the attacker?
11. **(E3)** Name one defence that takes away the attacker’s *feedback* instead of hardening the model, and the attacker’s answer to it.
12. **(Ethics)** The same code that measures robustness also helps a scammer. Which parts of what you built would you *not* publish, and why?

	Level 1 evasion at 12 words	Level 2 useful ASR	after normali- sation (homo- glyph attack, recall)
Victim			
Naive Bayes (counts)			
TF-IDF + logistic regression			
Nomic v2 MoE + LR (if available)			

Optional challenges

- **Universal suffix:** find a single 8-word suffix, on training spam only, that evades most of the test spam (`al.universal_trigger`). Which words does it contain?
- **Poisoning:** send spam with a rare trigger token, labelled *not spam*, into the training set (notebook 10, part A). Which model is more vulnerable, and why? Try a label-noise filter.
- **Judge:** use Nomic instead of MiniLM as the judge of meaning. Does the useful ASR change?
- **Paraphrase:** use the paraphrases of `datasets/spam_paraphrases.json` (made by Gemma 4 12B with the payload protected by placeholders) as one more operator (`al.op_paraphrase`).

Grading

Working implementations that pass the checks (50%), the plots and tables (25%), the answers to the questions (25%). Reference solution: `solution/guide_02_03.ipynb`.